

Support Ticket Triage Prototype

OCCA assignment | Phase 1 prototype | 9 October 2026

Architecture and why. Python/uv provides a reproducible CLI; Pydantic validates ticket, tool and result contracts. An explicit OpenAI GPT loop reads the whole conversation, invokes customer history and classic-RAG search, then returns urgency, product/issue/sentiment, routing, rationale and citations. Docker runs PostgreSQL/pgvector with repeatable migrations and persistent local storage. Source/version/mock metadata travels with retrieved passages. The small corpus uses exact cosine search (top five), avoiding unnecessary orchestration. Customer history and knowledge are synthetic, as allowed by the assignment. GraphRAG is deferred to the next phase.

Failures and safeguards. Six model requests and eight tool executions cap each ticket; provider timeout is 30 seconds, with one shared retry/correction allowance and no hidden SDK retries. Unknown tools, foreign customer lookup, invalid arguments/JSON/citations, tool failures or exhausted budgets produce explicit human-support fallback. Missing history permits disclosed uncertainty. Critical incidents and billing require human escalation; auto-response requires low urgency, retrieved evidence and a draft. Mock-based drafts are labeled demonstrations and never sent. Ticket/KB content is untrusted; tools are read-only and SQL is parameterized. Thai drafts remain Thai. Reported charges/outages stay unverified; no refunds, SLAs or unsupported features are promised. Prompt rules reduce hallucinations but cannot guarantee semantic truth.

Evidence and production evaluation. Offline tests cover all twelve original messages, real Docker retrieval, adapter HTTP contracts, three sample routes, invalid output and failures. The sample runner uses a deterministic demonstration model and fake lexical embeddings; it proves wiring, not GPT or multilingual semantic quality. No live provider calls ran because no key was configured. Before production, evaluate labeled English/Thai and mixed-language tickets for severity/routing accuracy, sentiment/extraction fidelity, recall@5/MRR, groundedness, false automation, prompt injection, latency, cost and fallback rate. Use held-out cases, repeated live runs and human review; add tenant isolation, approved real knowledge, observability and access controls before real customer use.